Deep Learning Architectures in Natural Language Understanding

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1. Introduction and Architectural Overview

Transformer architectures have redefined statistical language modeling by replacing recurrence with multi-head self-attention mechanisms. By enabling global context interaction across token positions, attention layers capture long-range semantic dependencies that recurrent networks historically degraded.

2. Empirical Latency and Memory Trade-Offs

The empirical performance across hardware profiles is recorded below:

Model Architecture | Context Window | Inference Latency (ms) | Memory Footprint (GB)

Llama-3-8B | 8192 | 18.4 | 16.2

Mistral-7B | 8192 | 16.1 | 14.5

Claude-3-Sonnet | 16384 | 42.0 | Cloud Managed

3. References and Citations

[1] Vaswani, A., et al. (2017). Attention is all you need. In NeurIPS (pp. 5998-6008).

[2] Devlin, J., et al. (2018). BERT: Pre-training of deep bidirectional transformers. arXiv:1810.04805.

[3] Brown, T., et al. (2020). Language models are few-shot learners. In NeurIPS (Vol. 33, pp. 1877-1901).